

Research Methodology in Public Administration

Proposal Writing and Report Writing

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Research Methodology

The Overall Strategy and Logic for Conducting Research

**Research Problem → Research Questions → Research
Methodology → Data Collection → Data Analysis**

What is Research Methodology?

Research methodology refers to the overall approach, strategy, logic, and rationale used to conduct a research study.

It explains:

- ▶ What will be studied?
- ▶ Why will it be studied?
- ▶ How will it be studied?
- ▶ What type of data will be required?
- ▶ How will the data be collected?
- ▶ How will the data be analyzed?

Research Methodology and Research Methods

Research Methodology

The overall **strategy, approach, and rationale** guiding the research.

It explains:

- ▶ Why a particular approach is selected
- ▶ How the study will be conducted
- ▶ The logic behind the methods

Research Methods

Specific **techniques and procedures** used to collect and analyze data.

Examples:

- ▶ Survey
- ▶ Interview
- ▶ Observation
- ▶ Experiment
- ▶ Statistical analysis

The Research Methodology Framework

A research methodology is not only about selecting a method.



Major Components of Research Methodology

A research methodology generally includes:

1. Research philosophy
2. Research approach
3. Research design
4. Research strategy
5. Population and sampling
6. Data collection methods
7. Measurement
8. Data analysis
9. Ethical considerations

Research Philosophy

A **research philosophy** refers to assumptions about:

- ▶ What reality is
- ▶ What can be known
- ▶ How knowledge can be acquired
- ▶ How research should be conducted

Research philosophy influences the selection of research approaches, methods, and interpretation of findings.

Major Research Philosophies

Some major philosophical approaches include:

- ▶ Positivism
- ▶ Post-positivism
- ▶ Interpretivism
- ▶ Critical theory
- ▶ Constructivism
- ▶ Pragmatism

Important

Different philosophical positions may lead researchers to ask different questions and use different methods.

Positivist Approach

The positivist approach generally emphasizes:

- ▶ Objective reality
- ▶ Observation and measurement
- ▶ Hypothesis testing
- ▶ Identification of relationships
- ▶ Explanation and prediction
- ▶ Quantitative data

Example

Does digital literacy significantly influence citizens' satisfaction with digital public services?

Interpretive Approach

The interpretive approach generally focuses on:

- ▶ Human experiences
- ▶ Meanings and interpretations
- ▶ Social context
- ▶ Individual perspectives
- ▶ Understanding social reality

Example

How do citizens experience and interpret digital public services?

Pragmatism

Pragmatism focuses on the research problem and practical consequences.

It allows researchers to use:

- ▶ Quantitative methods
- ▶ Qualitative methods
- ▶ Mixed methods

Key Principle

The choice of methods should be guided by the research problem and research questions.

Research Approaches

Research approaches provide the general direction for conducting research.

Quantitative Research

Focuses primarily on numerical data, measurement, and statistical analysis.

Qualitative Research

Focuses primarily on meanings, experiences, perspectives, and social processes.

Mixed Methods Research

Combines quantitative and qualitative approaches.

Quantitative Research

Quantitative research generally focuses on:

- ▶ Measurement
- ▶ Numerical data
- ▶ Variables
- ▶ Hypothesis testing
- ▶ Relationships between variables
- ▶ Statistical analysis

Example Question

What factors significantly influence citizens' satisfaction with digital public services?

Qualitative Research

Qualitative research generally focuses on:

- ▶ Meanings
- ▶ Experiences
- ▶ Perceptions
- ▶ Social processes
- ▶ Context
- ▶ Interpretation

Example Question

How do citizens experience challenges while accessing digital public services?

Mixed Methods Research

Mixed methods research combines quantitative and qualitative approaches within a single study.

Quantitative Data

- ▶ How many?
- ▶ How much?
- ▶ What relationship?



Combined with



Qualitative Data

- ▶ Why?
- ▶ How?
- ▶ What does it mean?

Research Design

A **research design** is the overall plan or blueprint for conducting a research study.

It specifies:

- ▶ What information is required
- ▶ From whom the information will be collected
- ▶ How the information will be collected
- ▶ When the information will be collected
- ▶ How the information will be analyzed

Common Types of Research Design

Some commonly used research designs include:

- ▶ Exploratory
- ▶ Descriptive
- ▶ Explanatory
- ▶ Experimental
- ▶ Case study
- ▶ Cross-sectional
- ▶ Longitudinal
- ▶ Comparative

Exploratory, Descriptive and Explanatory Research

Exploratory

Used when little is known about a phenomenon.

Purpose:

Explore and understand.

Descriptive

Describes characteristics of a population or phenomenon.

Purpose:

Describe what exists.

Explanatory

Examines relationships and possible causes.

Purpose:

Explain why or how.

Research Strategies

A research strategy provides a practical way of conducting the study.

Common research strategies include:

- ▶ Survey
- ▶ Case study
- ▶ Experiment
- ▶ Field research
- ▶ Ethnography
- ▶ Action research
- ▶ Historical research

Sources of Data

Research may use:

Primary Data

Data collected directly by the researcher.

Examples:

- ▶ Surveys
- ▶ Interviews
- ▶ Observation
- ▶ Focus groups

Secondary Data

Data previously collected by other individuals or organizations.

Examples:

- ▶ Government reports
- ▶ Census data
- ▶ Research reports
- ▶ Administrative records

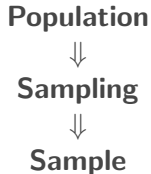
Population and Sampling

Population

The complete group of individuals, organizations, events, or units that the researcher is interested in studying.

Sample

A smaller group selected from the population for the purpose of conducting the study.



Sampling Approaches

Probability Sampling

Each unit has a known probability of selection.

Examples:

- ▶ Simple random
- ▶ Systematic
- ▶ Stratified
- ▶ Cluster

Non-Probability Sampling

The probability of selection is unknown.

Examples:

- ▶ Convenience
- ▶ Purposive
- ▶ Snowball
- ▶ Quota

Data Collection Methods

Common methods of collecting data include:

- ▶ Questionnaire
- ▶ Survey
- ▶ Interview
- ▶ Observation
- ▶ Focus group discussion
- ▶ Case study
- ▶ Document analysis
- ▶ Secondary data

Data Analysis

Data analysis involves organizing, processing, and interpreting data to answer the research questions.

Quantitative Analysis

- ▶ Descriptive statistics
- ▶ Correlation
- ▶ Regression
- ▶ Hypothesis testing

Qualitative Analysis

- ▶ Coding
- ▶ Categorization
- ▶ Thematic analysis
- ▶ Content analysis

How to Select an Appropriate Methodology

The choice of methodology should depend primarily on:

1. Research problem
2. Research questions
3. Research objectives
4. Nature of the phenomenon
5. Type of data required
6. Available resources
7. Time available
8. Researcher's skills
9. Ethical considerations

Matching Research Questions and Methodology

Research Question	Possible Approach	Possible Method
How many citizens use digital services?	Quantitative	Survey
What factors influence satisfaction?	Quantitative	Survey and statistical analysis
How do citizens experience digital services?	Qualitative	Interview
What factors influence satisfaction and why?	Mixed Methods	Survey + Interview

Research Methodology in a Proposal

A methodology section in a research proposal should generally explain:

1. Research philosophy or approach
2. Research design
3. Study area and population
4. Sampling method and sample size
5. Sources of data
6. Data collection methods
7. Data analysis techniques
8. Validity, reliability, or trustworthiness
9. Ethical considerations

Example of a Methodology Statement

Example

This study will adopt a **mixed methods approach** to examine the factors influencing citizens' satisfaction with digital public services. Quantitative data will be collected through a structured survey, while qualitative data will be collected through interviews with selected participants. Quantitative data will be analyzed using descriptive and inferential statistical techniques, whereas qualitative data will be analyzed thematically.

Every methodological choice should be justified.

Ethical Considerations

Research methodology should also address ethical issues.

- ▶ Informed consent
- ▶ Voluntary participation
- ▶ Privacy and confidentiality
- ▶ Anonymity
- ▶ Avoidance of harm
- ▶ Data protection
- ▶ Honesty and academic integrity

The Complete Research Methodology Logic



Key Takeaway

Research Methodology is more than selecting a method.

It provides the **overall logic and strategy** for conducting research.

Problem



Approach



Design



Methods

